

OPPONG KWABENA ADUTWUM

COMPUTER SCIENCE UNDERGRADUATE

Applied ML · Robotics · Embedded Intelligence · Full-Stack Development

CONTACT

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github.com/Oppong-py
oppong-py.github.io

TECHNICAL SKILLS

Programming Languages

- Python
- C++

Data & Analytics

- NumPy
- SQL
- Jupyter Notebooks

AI & ML Tools

- Scikit-learn
- Groq AI API
- Claude Code / Claude Cowork

Web & Backend

- React Native / Expo

Dev Tools & Design

- Git / GitHub
- VS Code
- Figma
- Linux (Kali)

Core Knowledge

- Machine Learning Fundamentals
- PID Control / Robotics
- Data Structures & Algorithms
- OSI & TCP/IP Networking

LANGUAGES

- English — Native
- French — Intermediate
- Twi — Native

CERTIFICATIONS

- Hashgraph Developer Course — Hedera (Jan 2026)
- Hedera Certified Foundation (Apr 2026)
- IPPF Competitor — Brewer Foundation & NYU (2024-25)
- WASSCE General Science — Prempeh College (Sep 2025)

GITHUB HIGHLIGHTS

- gds-hackathon-2026-building-age
- ml-specialization-coursera
- oppong-py.github.io

INTERESTS

- Mathematics
- Machine Learning & AI
- Robotics & Embedded Systems
- UI/UX & Visual Design
- Movies & Video Games
- Sports (Football)

MEMBERSHIPS

- Prempeh College Robotics Club (2023–2025)
- AmaliTech Coding Club (2025–Present)
- Microsoft User Group — KNUST (2025–Present)
- Claude Builder Club — KNUST (2025–Present)

PROFESSIONAL SUMMARY

First-year Computer Science student at Kwame Nkrumah University of Science and Technology with a strong interest in machine learning, robotics, and embedded systems. National Robotics Champion (Robofest Ghana 2025) and Regional Cybersecurity Champion (2024). Working through Andrew Ng's ML Specialization (Stanford/Coursera) — Course 1 of 3, notes published publicly.

EXPERIENCE

Machine Learning Trainee — ZapTek Research and Innovation Hub

Remote | May 2026

- Completed a 2-week introductory ML program covering data collection, preprocessing, model training and evaluation; currently in the second cohort.
- Cleaned datasets using Pandas and NumPy, handling missing values and feature scaling.
- Trained classification models with Scikit-learn and compared performance using accuracy, precision, recall, and F1-score.

Online Safety Analyst & Volunteer — Solutions for Life Initiative Ghana (SFLIG)

Ghana | Oct 2022 – Present

- Joined as volunteer (Oct 2022) for community support projects; promoted to Online Safety Analyst (Aug 2023), managing social media and website with focus on internet security.
- Runs speaking engagements with girls from orphanages and rural communities on online safety topics.

Co-Founder & Software Developer — Arethos Tech Initiatives

Kumasi, Ghana | 2026 – Present

- Co-founding and leading development of Arethos, a mental wellness app for KNUST students. Managing the technical roadmap, implementing AI-powered features, and coordinating a small dev team.
- Won 2nd Place at ACES CodeFest 2026 UI/UX Challenge. Redesigned the ACES KNUST website for mobile across all 9 pages as a team of 5, building a clickable Figma prototype.

EDUCATION

Kwame Nkrumah University of Science & Technology (KNUST)

B.Sc. Computer Science | Kumasi, Ghana | Expected 2029

- Relevant Coursework: Networking & Internet (OSI/TCP/IP, IPv4, subnetting), Calculus I, Probability & Statistics, C++ / CS & Math Programming

Prempeh College (Senior High School)

WASSCE — General Science | Kumasi, Ghana | Graduated 2025

PROJECTS

Autonomous Line-Following Robot

LEGO MINDSTORM EV3 · PID Control Algorithm

- Designed and hand-tuned a PID control algorithm for autonomous line-following; National Champion at Robofest Ghana 2025.
- Tuned the sensor-motor feedback loop by iteration — run, watch the oscillation, adjust, run again.

Class Attendance Monitoring System

React Native · Django · PostgreSQL · Bluetooth

- Built a biometric attendance platform and tested it with a class of 50 students.
- Built a Django REST API handling concurrent Bluetooth fingerprint scans, including the race conditions that come with them.

Solar-Powered Automated Water Dispenser

Arduino · Ultrasonic Sensors · C++ · Solar

- Built a solar-powered water dispensing system using dual ultrasonic sensors to detect container placement and fill level, automatically controlling a solenoid valve.
- Implemented debounce logic, buzzer feedback, LED status indicators, and off-grid solar power for deployment without reliable electricity.

Arethos — Mental Wellness App (In Development)

React Native · Expo SDK 54 · SQLite · Groq AI

- Cross-platform mental wellness app with AI-assisted journaling, mood tracking and adaptive reflections, built local-first with Groq AI.

HACKATHONS & COMPETITIONS

Ghana Data Science Regional Hackathon 2026 — Participant — Machine Learning (Sept 2026)

- Building-age classification from 40 years of Landsat imagery, transferred Madrid to Amsterdam; team of four. Found the supplied evaluation inflated by 0.047 macro-F1 through spatial leakage. github.com/Oppong-py/gds-hackathon-2026-building-age

ACES CodeFest 2026 — 2nd Place — UI/UX Design (July 2026)

- Redesigned the ACES KNUST website for mobile across all 9 pages as a team of 5. Built a high-fidelity clickable prototype in Figma with real-time collaboration, focusing on visual hierarchy, navigation, and user flows.

Robofest Ghana — National Champion — Robotics (2025)

- Built autonomous robots under timed conditions to complete designated tasks; designed and assembled within strict time limits.

National Cybersecurity Challenge — Regional Champion (2024)

- Competed in a knowledge-based quiz testing cybersecurity fundamentals, threat analysis, and security best practices.

LEADERSHIP & VOLUNTEERING

Club Leader — Prempeh College Cybersecurity Club (2023–2025)

Participant — International Public Policy Forum (IPPF) (2024)

Emmanuel Owusu Ansah — Senior Teacher, Mathematics, Prempeh College +233 54 538 3894

Kwadwo Gyan-Darkwa — Physics Teacher, Prempeh College +233 244 940 819

Sally Anokye-Yeboah (Mrs) — Executive Director, SFLIG +233 559 1864